

(A405311) DEVOPS**B. Tech (CSE) V Semester**

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UNIT-I

Introduction to DevOps:

Introduction, Agile development model, DevOps and ITIL. DevOps process and Continuous Delivery, Release management, Scrum, Kanban, delivery pipeline, identifying bottlenecks.

UNIT-II**Software development models and DevOps:**

DevOps Lifecycle for Business Agility, DevOps, and Continuous Testing. DevOps influence on Architecture: Introducing software architecture, The monolithic scenario, Architecture rules of thumb, The separation of concerns, Handling database migrations, Micro services and the data tier, DevOps, architecture, and resilience.

UNIT-III**Introduction to project management:**

The need for source code control, the history of source code management, Roles and code, source code management system and migrations, shared authentication, Hosted Git servers, Different Git server implementations, Docker intermission, Gerrit, The pull request model, GitLab.

UNIT-IV**Integrating the system:**

Build systems, Jenkins build server, managing build dependencies, Jenkins plugins, and file system layout, The host server, build slaves, Software on the host, Triggers, Job chaining and build pipelines, build servers and infrastructure as code, Building by dependency order, Build phases, Alternative build servers, Collating quality measures.

UNIT-V**Testing Tools and Deployment:**

Various types of testing, Automation of testing Pros and cons, Selenium - Introduction, Selenium features, JavaScript testing, Testing backend integration points, Test-driven development, REPL-driven development. Deployment of the system: Deployment systems, Virtualization stacks, code execution at the client, Puppet master and agents, Ansible, Deployment tools: Chef, Salt Stack and Docker.

TEXTBOOKS

1. Joakim Verona., Practical DevOps, Packt Publishing, 2016.

REFERENCE BOOKS

1. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint. Wiley publications.
2. Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley

COURSE OUTCOMES:

On completion of the course students will be able to

1. Identify components of Devops environment.
2. Describe Software development models and architectures of DevOps.
3. Apply various Source control systems for different project managements
4. Integrate the Systems with various build Servers for various software application practices.
5. Choose an appropriate testing tool and deployment model for the project

CO-PO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1			3				1			1
CO2	1	2	3		3				3			2
CO3	1	2	3		3				3			2
CO4	1	2	3		3				3			2
CO5	1	2	3		3				1			2

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